

RARD: Rationale-First Blockwise Autoregressive Diffusion in Rationale-Dominated Graph Generation

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Abstract

Graph generation underlies many critical applications, from social network modeling to knowledge graph reasoning. Across these diverse domains, many graphs are *rationale-dominated*: a small, semantically meaningful subgraph determines the property of interest, while the remaining edges contribute largely noisy variation. Despite the significance of this inherent structure, existing generative methods often fail to preserve these task-critical substructures. We introduce RARD (Rationale-first blockwise AutoRegressive Diffusion), a topology-guided framework that learns to *separate and prioritize* the rationale. RARD employs a persistent-homology-based learning objective to discover an optimal graph filtration, an edge ordering that explicitly separates rationale from noise. Building upon this learned filtration, RARD generates graphs blockwise: it adds filtration-aligned blocks autoregressively and refines each new block with a shared local discrete diffusion module, ensuring the rationale appears early while peripheral structure is added later. We provide theoretical analysis showing that maximizing the topological gap yields rationale-first ordering and collapses to a two-level filtration. Comprehensive experiments across seven benchmarks demonstrate that RARD achieves state-of-the-art performance on widely used metrics.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Graph generation; graph neural network; graph filtration learning

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1 Introduction

Graph generation lies at the heart of numerous web-scale and scientific applications, ranging from social network analysis [3] and knowledge graph reasoning [22], to online fraud detection [27] and drug discovery [20, 32]. Central to these applications is a common yet often overlooked characteristic: graphs frequently exhibit an inherent hierarchical structure dominated by a semantically meaningful subgraph—the *rationale*—that governs key properties, with the remaining connections serving merely as noise or superficial decoration. For example, social graphs contain tightly knit *core communities* that drive interactions, surrounded by loosely connected peripheries [25]; knowledge-graph reasoning frequently relies on minimal relational chains that form the logical evidence for answering queries [29]; financial transaction graphs feature *fraud-detection motifs* such as short cyclic money-flow patterns; and in molecular design, *pharmacophores* encode the functional substructures crucial for bioactivity [28]. We collectively refer to these as *rationale-dominated graphs*.

Despite the practical significance of rationale-dominated graphs, existing graph generation algorithms focus on optimizing global realism while remaining agnostic to structural faithfulness. Autoregressive (AR) models [21, 40], generate nodes and edges sequentially to maximize likelihood under heuristic or arbitrary ordering, treating all edges uniformly. As a result, AR models primarily match global statistics—such as degree distributions or clustering coefficients—but often fail to reconstruct critical substructures, including cycles, motifs, or communities. Diffusion-based models [14, 26], achieve high fidelity by iteratively denoising, but often lack mechanisms to distinguish or preserve task-relevant subgraphs. Consequently, topological motifs are frequently diluted, delayed, or broken (e.g., short transaction cycles that indicate fraud). Thus, a fundamental question emerges: *Can we design a principled hybrid generation framework that (i) explicitly discovers and prioritizes the rationale subgraph, and (ii) refines residual noise with diffusion—thereby preserving task-critical motifs (e.g., cycles) while achieving both interpretability and high sample quality?*

We introduce a topology-guided framework that learns to *separate and prioritize* the rationale. Using *persistent homology* (PH) [6], we discover an optimal *graph filtration*—an edge ordering that

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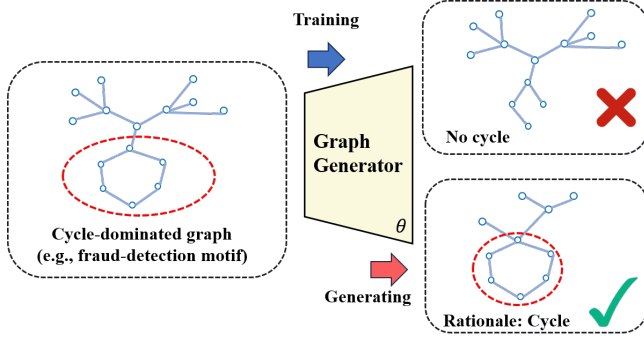


Figure 1: Preserving the rationale in graph generation. Left: a cycle-dominated training graph (e.g., fraud-detection motif); the dashed red ellipse marks the cycle rationale. Right: a conventional generator produces a plausible sample but omits the cycle (top), whereas our rationale-aware approach preserves the cycle (bottom).

maximizes a *topological gap* (measured via bottleneck distance [9]) between early (rationale-dominated) and late (noise-dominated) stages. This learned, rationale-first filtration then drives a *blockwise autoregressive diffusion* generator that produces core subgraphs first and adds peripheral edges later, combining the interpretability and efficiency of AR with the sample quality of diffusion. The framework is shown in Figure 3. Our main contributions are:

- We propose a differentiable PH-based objective that learns a rationale–noise hierarchy by maximizing the topological gap between sub-filtrations.
- We introduce a filtration-aligned, blockwise autoregressive diffusion process that preserves critical motifs while maintaining diffusion-level realism.
- Comprehensive experiments on seven benchmarks (social and molecular) demonstrate consistent state-of-the-art performance and robust rationale preservation.

2 Related Work

Graph Generation . Autoregressive (AR) models construct graphs incrementally by sequentially adding nodes and edges, naturally aligning with the discrete structure of graphs. However, such methods inherently face the challenge of lacking a canonical generation order. GRAPHARN [40] addresses this by aligning permutations with breadth-first search (BFS) orderings, creating a many-to-one mapping. GRAN [21] attempts to mitigate order bias by marginalizing over multiple canonical node orders, such as descending node degrees, BFS trees rooted at high-degree nodes, and k-core decompositions. [5] further eliminate heuristic ordering entirely, modeling node orders themselves as latent variables with an additional AR model, thus estimating marginalized probabilities jointly over graph structures and their orders. Despite these advances, all autoregressive methods fundamentally rely on artificially imposed node orderings.

Recent works have adapted diffusion models to graph generation tasks under both continuous and discrete formulations. On the continuous side, EDP-GNN [26] introduces score-based methods

for graphs by treating adjacency matrices and node attributes as continuous-valued data, and utilizes stochastic differential equations (SDEs) for the diffusion process. GDSS [14] generalizes this framework by jointly diffusing node and edge features. However, modeling exchangeable distributions via purely permutation equivariant architectures remains challenging, prompting permutation-sensitive designs like SWINGNN [38] to improve continuous-state score matching. Despite progress, these continuous approaches inherently neglect the discrete structure of graphs. To address this, discrete diffusion models have emerged, with DiGRESS [33] being the first to explicitly employ discrete denoising diffusion for graph generation, achieving significant performance gains but relying on structural priors and domain-specific features. DISCO [37] further formulates graph diffusion in a discrete-state continuous-time setting, preserving the discrete nature of graphs while enabling flexible sampling efficiency trade-offs. Nonetheless, diffusion models typically require multiple denoising steps, limiting sampling speed. To alleviate this, recent efforts integrate diffusion with autoregressive (AR) approaches. GRAPHARM [17] proposes an autoregressive diffusion model (ADM) that sequentially transitions nodes and their incident edges to absorbing states following a random node order. PARD [42] combines diffusion and AR by partitioning the graph into blocks, applying diffusion within each block, and using an AR procedure between blocks.

Graph Filtration. Filtration is widely adopted in persistent homology [8] to capture structural features of geometric data at multiple resolutions. In previous work, graph filtration mainly focused on extracting informative graph representations [44] suitable for downstream tasks, such as classification [41] or prediction [39]. Recently, its applications have expanded significantly, including graph explainability [36], graph generation [18], and graph evaluation [31].

3 Preliminaries

3.1 Rationale-Dominated Graph

We study the generative process of graphs $G = (V, E)$ that admit a decomposition into a task-relevant rationale subgraph and a task-irrelevant noise subgraph. We denote this decomposition as $G = G_X \sqcup G_E$, where G_X and G_E respectively represent the rationale and noise subgraphs. Specifically, we assume an edge-level disjoint union $E = E_X \sqcup E_E$, where $E_X \cap E_E = \emptyset$. The corresponding subgraphs are defined as $G_X = (V_X, E_X)$ with $V_X = \{v \in V \mid \exists e \in E_X \text{ such that } v \in e\}$, and $G_E = (V_E, E_E)$ with $V_E = \{v \in V \mid \exists e \in E_E \text{ such that } v \in e\}$. Note that vertex sets are not required to be disjoint. Intuitively, the subgraph G_X , which is typically small and semantically meaningful, captures the task-relevant structure and determines the label y_G , while the complementary subgraph G_E encodes task-irrelevant or noisy information. This decomposition naturally induces a canonical autoregressive generation order: $\emptyset \rightarrow G_X \rightarrow G_X \cup G_E = G$ where the rationale subgraph G_X is generated first, followed by the addition of spurious or task-irrelevant G_E .

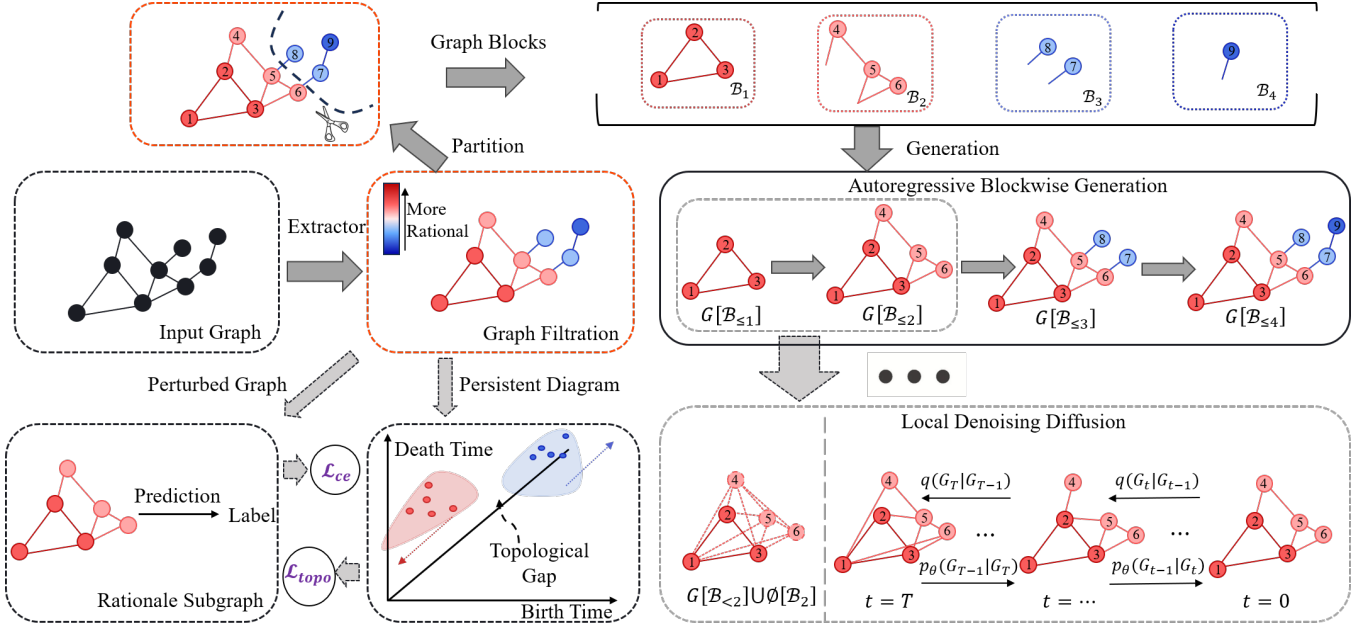


Figure 2: Framework of RARD. An extractor predicts edge rationality scores and induces a rationale-first filtration (red = rationale). Persistence diagrams from this filtration drive a topological-gap objective (with label supervision) that separates rationale from noise. The filtration partitions the graph into ordered blocks; at generation time, blocks are added autoregressively and each new block is refined by a shared local discrete denoising diffusion, so rationale appears early and peripheral structure is added later.

3.2 Graph Generation

Autoregressive Graph Generation. Autoregressive (AR) models define a factorized likelihood by decomposing a joint distribution into a product of conditional terms. Let G be a target graph and let $\{G_t\}_{t=0}^T$ be a sequence where $G = G_T$ and each transition $G_{t-1} \rightarrow G_t$ adds a small, well-defined structure (e.g., one node and its incident edges to previously introduced nodes, or a block of adjacency entries). An AR model then factorizes

$$\log p_\theta(G) = \sum_{t=1}^T \log p_\theta(G_t | G_{<t}),$$

with $G_{<t} = \{G_0, \dots, G_{t-1}\}$. Each conditional distribution $p_\theta(G_t | G_{<t})$ is parameterized by a neural network and trained via next-step prediction, enabling efficient learning through next-node or next-edge objectives.

Discrete Denoising Diffusion on Graph. Following D3PM [2] and its application to graph generation as in DiGress [33], a graph $G_0 = G$ is modeled with a discrete forward noising Markov chain of T step and a learned reverse chain, where node tokens v_t^i and edge tokens e_t^{ij} are categorical at step t . The forward process factorizes independently over tokens,

$$q(G_t | G_{t-1}) = \prod_{i \in V} q(v_t^i | v_{t-1}^i) \prod_{i,j \in V} q(e_t^{ij} | e_{t-1}^{ij}),$$

while parameterization of reverse denoising process conditions on the whole noisy graph at step t ,

$$p_\theta(G_{t-1} | G_t) = \prod_{i \in V} p_\theta(v_{t-1}^i | G_t) \prod_{i,j \in V} p_\theta(e_{t-1}^{ij} | G_t).$$

Then the Variational Lower Bound (VLB) of $\log p_\theta(G_0)$ can be written as:

$$\log p_\theta(G_0) \geq -\mathcal{L}_1(\theta) - \mathcal{L}_{\text{prior}} - \sum_{t=2}^T \mathcal{L}_t(\theta), \quad (1)$$

where the terms are

$$\mathcal{L}_1(\theta) = -\mathbb{E}_{q(G_1 | G_0)} [\log p_\theta(G_0 | G_1)],$$

$$\mathcal{L}_{\text{prior}} = \text{KL}(q(G_T | G_0) \| p_\theta(G_T)) \approx 0,$$

$$\mathcal{L}_t(\theta) = \mathbb{E}_{q(G_t | G_0)} [\text{KL}(q(G_{t-1} | G_t, G_0) \| p_\theta(G_{t-1} | G_t))].$$

Instead of parameterizing $p_\theta(G_{t-1} | G_t)$ directly, we follow [43] to predict $p_\theta(G_0 | G_t)$ and derive the one-step reverse process from it. We solely use the cross entropy loss between $q(G_t | G_0)$ and $p_\theta(G_0 | G_t)$ to quantify reconstruction quality:

$$\mathcal{L}_t^{\text{CE}}(\theta) = -\mathbb{E}_{q(G_t | G_0)} \left[\sum_i \log p_\theta(v_0^i | G_t) + \sum_{i,j \in V} \log p_\theta(e_0^{ij} | G_t) \right]$$

3.3 Computational Topology for Graphs

Graph Filtration. Let $G = (V, E)$ be a graph where V is the vertex set and $E \subset V \times V$ the edge set. Graphs in our case are undirected and simple, where there are no multiple edges between any two

vertices. Equipping G with a normalized edge weight map $f : E \rightarrow [0, 1]$, we obtain for each threshold $\lambda \in [0, 1]$ the sublevel edge set $E^\lambda := \{e \in E | f(e) \leq \lambda\}$ and the sublevel graph $G^\lambda := (V^\lambda, E^\lambda)$ (V^λ induced by E^λ). Because $E^{\lambda_1} \subseteq E^{\lambda_2}$ whenever $\lambda_1 \leq \lambda_2$, the family $\{G^\lambda\}_{\lambda \in [0, 1]}$ forms an ascending sequence of subgraphs of G : $\emptyset = G^{\lambda_0} \subseteq G^{\lambda_1} \subseteq \dots \subseteq G^{\lambda_t} = G$ where $0 = \lambda_0 < \lambda_1 < \dots < \lambda_t = 1$ are the distinct intervals. This sequence of nested subgraphs is called graph filtration $\mathcal{F}(G)$.

Persistent Homology. Persistent homology (PH) tracks the evolution of homological features in a topological space as the sublevel sets of a given function grow. In the case of graphs, these features correspond to the appearance and disappearance of connected components and cycles, encoded as birth-death intervals determined by the inclusion of vertices and edges during the filtration process. We treat vertices as present once any incident edge enters, so births and deaths are expressed entirely in terms of edge weights.

Zeroth homology H_0 . A new connected component is born at the smallest threshold at which one of its edges appears; with the convention that isolated vertices are present at the start, we may take all H_0 births to occur at $\lambda = 0$. When an edge e with filtration value $f(e)$ joins two previously disjoint components, the younger component (the one whose most recent birth time is larger) dies at $f(e)$. Thus each H_0 bar is a pair $(b, d) = (0, f(e_{\text{merge}}))$, and the multiset of these pairs constitutes the 0-dimensional barcode.

First homology H_1 . The first homology H_1 is the vector space defined on the set of all cycles in G , with addition operation defined as the symmetric difference of cycles, which is well studied in graph theory [7, 12, 15]. In a standard graph filtration, a cycle created by adding an edge need not die because no higher-dimensional simplex appears to fill it, leading to infinite bars. To obtain finite lifetimes, we adopt *extended persistent homology* [6]: we pair the usual sublevel filtration with the complementary superlevel filtration so that every cycle is eventually "killed" when the filtration switches to removing edges in decreasing order. Concretely, for each cycle $C \subseteq E$, let $b(C) = \min_{e \in C} f(e)$ be the filtration value of the earliest edge of the cycle and $d(C) = \max_{e \in C} f(e)$ be that of the latest edge. The interval $(b(C), d(C))$ is then recorded as a 1-dimensional bar. In this way, both the birth and death times of H_1 features are determined by edge weights, and no bar persists to infinity.

Collecting all birth–death pairs (b, d) from H_0 and H_1 yields the barcode $\mathcal{B} = \{(b, d)\}$ and equivalently the persistence diagrams $\mathcal{D}$. Each bar (b, d) has persistence $|d - b|$, measuring how long the corresponding topological feature survives across the edge-ordered growth of the graph. Because the diagrams $\mathcal{D}_0$ and $\mathcal{D}_1$ serve as stable topological signatures of the edge-weighted graph, one can compare two such graphs by computing distances between the persistence diagrams induced by their respective filtrations. Metrics such as the bottleneck distance d_B [9] translate perturbations in edge weights into provably small changes in the resulting topological summaries, thereby ensuring stability under bounded noise or structural variation.

4 Methodology

We address graph learning and generation in settings where each rationale-dominated graph $G = (V, E)$ admits a latent split $G = G_X \sqcup G_\epsilon$ and a learned graph filtration function $f_\alpha : E \rightarrow [0, 1]$ can

reveal this structure. The motivation is that ideal filtration should order edges so that label-determining ones appear early and form a topologically compact part, while later edges are predominantly noisy. Once such an ordering is available, it is natural to align the generative mechanism with it rather than treating all edges symmetrically.

4.1 Rationale-First Filtration Learning

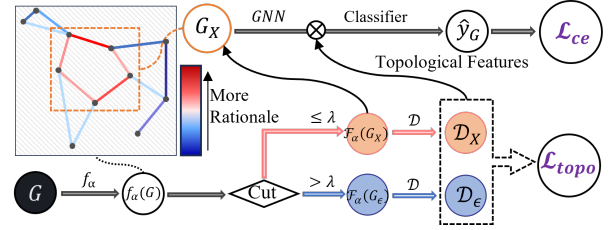


Figure 3: Graph filtration's high-level architecture. Starts from a filtration function f_α to obtain a scalar edge embedding $f_\alpha(G)$. After passing a predefined hard cut, two graph filtration $\mathcal{F}(G_X)$ and $\mathcal{F}(G_\epsilon)$ are induced for computing topological features and their topological loss. The rationale subgraph G_X , together with topological features, jointly form the representation for predicting the graph label.

Let $G = (A, X)$ be a graph with n nodes and m edges, where $A \in \{0, 1\}^{n \times n}$ denotes the adjacency matrix and $X \in \mathbb{R}^{n \times d}$ is the node feature matrix with feature dimension d . In this work, we focus on *rationale-first filtration learning* for the task of graph classification. Given a collection of labeled graphs $\{(G_i, y_i)\}_{i=1}^N$ where $G_i \in \mathcal{G}$ and $y_i \in \mathcal{Y} = \mathbb{R}^c$ for c classes, our objective is to train a subgraph-extraction network $f_c \circ g$ that first identifies the task-relevant (*rationale*) subgraph and its task-irrelevant (*noise*) complement, and then predicts the label.

Formally, the extractor $g : \mathcal{G} \rightarrow \mathcal{G}_X \sqcup \mathcal{G}_\epsilon$ partitions an input graph into a rationale subgraph G_X and a noise subgraph G_ϵ , while the classifier $f_c : \mathcal{G}_X \rightarrow \mathcal{Y}$ predicts the label based on G_X . Here, $\mathcal{G}$, $\mathcal{G}_X$, $\mathcal{G}_\epsilon$, and $\mathcal{Y}$ denote the spaces of input graphs, rationale subgraphs, noise subgraphs, and labels, respectively.

During subgraph extraction, g adopts an attention mechanism to estimate the importance of each edge. Given node representations $\{H_i \in \mathbb{R}^h\}_{i=1}^n$ obtained via message passing, the extractor computes an **edge rationality score** $\alpha_e = \sigma(a([H_u, H_v]))$ for each edge $e = (u, v)$, where $a(\cdot)$ is a learnable function, $\sigma(\cdot)$ is the sigmoid activation, and $f_\alpha : \mathbb{R}^{2h} \rightarrow \mathbb{R}_+$ is referred to as the **Rationality Score Function**. The resulting attention vector $\alpha = [\alpha_1, \dots, \alpha_m]^\top$ defines an edge-wise probability distribution over subgraph importance. To obtain a discrete subgraph mask while maintaining differentiability, we sample $p_e \sim \text{Bernoulli}(\alpha_e)$ and apply the Gumbel-Softmax reparameterization trick [13] to ensure gradients are computable with respect to α_e .

Rationale-First Graph Filtration. Equipped with the learned rationality score function f_α , we define, for each graph G , a *rationale-first graph filtration* $\mathcal{F}_\alpha(G)$ that orders edges in **descending** order

of their rationality scores. For simplicity, we omit the subscript α when the context is clear.

For any threshold $\lambda \in [0, 1]$, let $\mathcal{F}(G^{\leq \lambda})$ denote the *sub-filtration* comprising edges whose scores satisfy $f(e) \geq 1 - \lambda$, which corresponds to the *high-importance* (rationale) structure. Conversely, we denote by $\mathcal{F}(G^{\geq \lambda})$ the complementary part containing edges with $f(e) < 1 - \lambda$, representing the *low-importance* (noise) structure. Intuitively, the filtration $\mathcal{F}_\alpha(G)$ induces an ordered sequence of subgraphs that progressively reconstruct the graph from its most essential rationale edges to its least informative noise edges. This filtration provides a principled foundation for persistent-homology-based analysis and later guides the blockwise autoregressive generation process.

Key Observation. There exists a global threshold $\lambda \in [0, 1]$ such that the distributions of the topological summaries $\mathbb{P}(\mathcal{D} \circ \mathcal{F}(G^{\leq \lambda}))$ and $\mathbb{P}(\mathcal{D} \circ \mathcal{F}(G^{\geq \lambda}))$ differ significantly when evaluated under the persistent diagram operator $\mathcal{D}$.

Remark. This observation arises from the fundamentally different topological behaviors of the rationale and noise subgraphs. The rationale subgraph G_X tends to exhibit stable and frequently recurring topological patterns across instances. In contrast, the noise subgraph G_ϵ typically comprises disordered or randomly structured edges, whose topological features are highly variable.

Topological Gap. To quantify this difference, we define the *topological gap* d_{topo} between the early and late parts of a filtration as the 1-Wasserstein distance between their respective persistence-diagram distributions. Let $\mathcal{D}_X = \mathbb{P}(\mathcal{D} \circ \mathcal{F}(G^{\leq \lambda}))$ and $\mathcal{D}_\epsilon = \mathbb{P}(\mathcal{D} \circ \mathcal{F}(G^{\geq \lambda}))$. Then, under the bottleneck distance d_B , we define

$$d_{\text{topo}}(\mathcal{D}_X, \mathcal{D}_\epsilon) := \inf_{\pi \in \Pi(\mathcal{D}_X, \mathcal{D}_\epsilon)} \mathbb{E}_{(P, Q) \sim \pi} [d_B(P, Q)], \quad (2)$$

where $\Pi(\mathcal{D}_X, \mathcal{D}_\epsilon)$ denotes the set of all admissible couplings between the two distributions.

We next establish theoretical conditions under which this topological gap is uniquely maximized by the ground-truth filtration, ensuring that rationale edges consistently appear *earlier* and noise edges *later* in the learned ordering.

THEOREM 4.1 (APPROXIMATE RATIONALE-FIRST OPTIMALITY). Let $G = (V, E)$ admit a decomposition $E = E_X \sqcup E_\epsilon$ into rationale and noise edges. Let $\mathcal{F}$ be the edge-induced sublevel filtration and write $\mathcal{D}(G^{\leq \lambda})$ and $\mathcal{D}(G^{\geq \lambda})$ for the persistence diagrams (H_0/H_1) computed on $G^{\leq \lambda}$ and $G^{\geq \lambda}$. Assume:

(1) **Approximate conditional sufficiency.** The label depends primarily on the rationale: $p(y | E_X, E_\epsilon) = p(y | E_X) + \delta_{\text{suf}}$, with $|\delta_{\text{suf}}| \leq \epsilon_{\text{suf}}$.

(2) **Topological separability.** There exists $\lambda^* \in [0, 1]$ such that

$$d_{\text{topo}}(\mathcal{D}(G^{\leq \lambda^*}), \mathcal{D}(G^{\geq \lambda^*})) \geq \Delta > 0.$$

(3) **Expressive, Lipschitz filtration family.** The learned filtration $f_\alpha : E \rightarrow [0, 1]$ is realized by a Lipschitz neural network on a compact edge-feature domain and is a universal approximator for continuous monotone edge-score maps.

(4) **Stable PH vectorization.** There exists a diagram embedding $\psi : X \rightarrow \mathbb{R}^k$ that is L_ψ -Lipschitz with respect to the bottleneck distance d_B . Let $T_X = \psi(\mathcal{D}(G^{\leq \lambda^*}))$, $T_\epsilon = \psi(\mathcal{D}(G^{\geq \lambda^*}))$.

Let $\hat{f}_\alpha$ be an empirical maximizer (up to optimization error ϵ_{opt}) of the surrogate objective $\|\mathbb{E}[T_X] - \mathbb{E}[T_\epsilon]\|$, and let $\epsilon_{\text{gen}}(N)$ bound the generalization gap from a sample of N graphs. Then:

(1) **Approximate edge-score ordering.** There exists $\kappa > 0$ (depending only on ψ and the diagram family) such that

$$\mathbb{E}[f_\alpha(e_X) - f_\alpha(e_\epsilon)] \geq \kappa \Delta - (\epsilon_{\text{opt}} + \epsilon_{\text{gen}}(N) + \epsilon_{\text{suf}}).$$

(2) **Binary convergence in probability.** As $\epsilon_{\text{opt}}, \epsilon_{\text{gen}}(N), \epsilon_{\text{suf}} \rightarrow 0$, the learned filtration converges in probability to a two-level map:

$$f_\alpha(e) \xrightarrow{p} \begin{cases} c_X, & e \in E_X, \\ c_\epsilon, & e \in E_\epsilon, \end{cases} \quad \text{with } c_X > c_\epsilon,$$

up to any strictly increasing reparameterization of $[0, 1]$.

Theorem 4.1 formalizes the learning guarantee behind our rationale-first filtration. Under mild assumptions, maximizing the PH-based *topological gap* not only enlarges the separation between the persistence diagrams of early (rationale-dominated) and late (noise-dominated) subgraphs, but also enforces an ordering on edge scores themselves. This separation of rationale edges directly enables the model to learn the underlying rationale distribution.

Realization of d_{topo} . In practice, we approximate d_{topo} using a stable and tractable surrogate. By the Kantorovich–Rubinstein duality [34], the topological gap $d_{\text{topo}}(\mathcal{D}_X, \mathcal{D}_\epsilon)$ bounds the maximal expected difference of any 1-Lipschitz functional ψ over persistence diagrams sampled from the two distributions:

$$\sup_{\psi \in \Psi} |\mathbb{E}_{P \sim \mathcal{D}_X} [\psi(P)] - \mathbb{E}_{Q \sim \mathcal{D}_\epsilon} [\psi(Q)]| \leq d_{\text{topo}}(\mathcal{D}_X, \mathcal{D}_\epsilon).$$

Following this formulation, we instantiate a differentiable and stable *persistent-homology (PH) vectorization* ψ [11] to map each persistence diagram into a finite-dimensional Euclidean space. Specifically, for a given rationality score function f_α , we first compute the corresponding persistence diagrams $\mathcal{D}_X$ and $\mathcal{D}_\epsilon$, then obtain their topological embeddings

$$T_X = \psi(\mathcal{D}_X), \quad T_\epsilon = \psi(\mathcal{D}_\epsilon),$$

which serve as vectorized representations of the rationale- and noise-dominated filtrations, respectively. The magnitude of their separation, $\|T_X - T_\epsilon\|$, provides a computable lower bound of d_{topo} . We further concatenate these two vectors to form a global topological descriptor $T_G = [T_X \| T_\epsilon]$, and apply the learned edge weights α to perturb the adjacency matrix, yielding $\tilde{G} = \alpha \odot A$. Finally, the classifier f_c jointly consumes the perturbed graph $\tilde{G}$ and the topology descriptor T_G to predict the label. The overall training objective integrates label supervision with topological separation as:

$$\min_{f_\alpha, f_c, \psi} \mathbb{E}_{G \sim \mathcal{G}} [L_{\text{ce}}(f_c(\tilde{G}, T_G), y) - \beta d_{\text{topo}}(T_X, T_\epsilon; f_\alpha, \psi)],$$

where $\beta > 0$ balances classification fidelity and topological discriminability.

4.2 Filtration-guided Blockwise Autoregressive Diffusion

Building upon the rationale-first filtration introduced in the previous subsection, we now describe how to construct a *blockwise autoregressive diffusion model* that decomposes graph generation

into a sequence of semantically meaningful substructures, or *blocks*, ordered along the learned filtration.

Overview. The rationale-first filtration naturally induces a two-level *partial order* on the graph through the filtration function f_α . While rationale edges consistently precede noise edges, edges within the same level may still vary in their scores or structural roles. Hence, edges belonging to the same level are mutually incomparable and admit no canonical internal order. A practical strategy is therefore to partition each level uniformly into smaller, approximately homogeneous blocks, enabling autoregressive modeling at a coarser, interpretable granularity.

Obtain Graph Blocks Induced by Filtration. Let the filtration function given by $f_\alpha(e) \in [0, 1]$. (i) We first threshold the edges at a fixed level τ (default as 0.6) to obtain $E_X = \{e \in E \mid f_\alpha(e) \geq \tau\}$ and $E_\varepsilon = E \setminus E_X$. Let $\text{Inc}(E') := \{v \in V \mid \exists e \in E', v \in e\}$. Then set $V_X = \text{Inc}(E_X)$ and $V_\varepsilon = \text{Inc}(E_\varepsilon) \setminus V_X$. This yields two coarse blocks $B^{(0)} = \{V_X, V_\varepsilon\}$. (ii) To refine each coarse block, we define a node-level score by averaging incoming edge scores:

$$s(v) = \frac{1}{\deg(v)} \sum_{(u,v) \in E} f_\alpha(u, v), \quad (3)$$

For V_X , we sort its vertices in descending order of $s(\cdot)$ and split the list uniformly into n sub-blocks $\{B_1^X, \dots, B_n^X\}$. For V_ε , we perform the same procedure but split into m sub-blocks $\{B_1^\varepsilon, \dots, B_m^\varepsilon\}$. (iii) Concatenating all refined sub-blocks gives the final block sequence $\mathcal{B} = \{B_1^X, \dots, B_n^X, B_{n+1}^\varepsilon, \dots, B_{n+m}^\varepsilon\}$. Let $B_{< i} := \bigcup_{j=1}^{i-1} B_j$ in order, then we can decomposes the joint probability of a graph G into(drop the superscript)

$$p_\theta(G) = \prod_{i=1}^{n+m} p_\theta(G[B_i] \mid G[B_{< i}]), \quad (4)$$

where $G[B_i]$ denotes the induced subgraph of G that contains precisely the vertices in B_i and the edges between them, and $G[B_{< i}]$ denotes the induced subgraph of G that contains precisely the vertices in $\bigcup_{j < i} B_j$ and the edges between them. Because each conditional probability is defined over a subset of nodes and edges conditioned on prior blocks, it is considerably simpler to model than the full joint distribution. This conditional probability is further decomposed into the probability of block size and the probability of block enlargement.

$$p_\theta(|B_i| \mid G[B_{< i}]) \prod_{x \in G[B_i]} p_\theta(x \mid G[B_{< i}] \cup \emptyset[B_i]), \quad (5)$$

where x denotes any node and edge in $G[B_i]$, $\emptyset$ denotes an empty graph to be enlarged ($|\cdot|$ denotes the size). Hence $G[B_{< i}] \cup \emptyset[B_i]$ means that we need to enlarge $G[B_{< i}]$ with same empty nodes or edges to the same size as $G[B_{\leq i}]$. To achieve this, for a graph with n nodes, we pad the graph with virtual nodes and edges to the size $2n$, such that the next part with nodes ranging from $n \sim 2n$ contains the predicted blocks.

After padding the graph to size $2n$, we apply the second part of Eq. 5 to perform blockwise denoising diffusion within the newly allocated region. Similar to the discrete denoising framework in Section 3.2, we treat node and edge features in the target block B_i as categorical variables and model the forward process as a Markov

chain that gradually adds discrete noise. During training, noise is introduced only to the node and edge tokens within B_i , while the tokens in previously generated blocks $B_{< i}$ remain unchanged. This localized noising strategy prevents interference with earlier predictions and enables the model to focus its reconstruction capacity on the current block, facilitating more stable and accurate blockwise generation. The detailed procedure is provided in Algorithm 1 in the Appendix.

To model the generative process along this blockwise decomposition, we adopt a shared local discrete denoising diffusion mechanism that operates over a series of blockwise enlargements. At each step, the model first predict a new block size conditioned on previously generated blocks, and then learn to refine the real block nodes and edges in diffusion. This design enables a gradual and structurally coherent transformation of the graph, where the diffusion dynamics remain consistent across blocks. We defer the technical implementation details to the Appendix B.

By aligning the generation process with the underlying filtration, it ensures that core substructures are generated early and that noise is progressively incorporated in a controlled and learnable manner. We defer the complexity analysis of RARD to the Appendix F.

5 Experiment

We evaluate our method on seven standard benchmark datasets with different rationale subgraphs and structural properties, including both synthetic and real graph generation. We will first provide an introduction to the datasets, baselines, and setup of the experiments.

5.1 Experiment Settings

Datasets. **BA-2Motifs** [23] is a synthetic binary-labeled graphs: house motifs and cycle motifs give class labels and thus are regarded as ground-truth rationale. **BA-HouseAndGrid** [4] is a synthetic binary-labeled graphs: label 1 when both house and grid motifs appear, label 0 otherwise. The house and grid(3×3) motifs are serve as ground-truth rationale because they do not have overlapping structures. **SPMotif** [35] is a synthetic three-class graphs: each class embeds a distinct motif as ground-truth rationale, with spurious correlations between the rest graph component and the labels. The degree of such correlation is controlled by $b = 0.5$. **Benzene** [30] comprises 12,000 molecular graphs and the carbon atoms in these benzene rings serve as the ground-truth rationale. **Mutag** [16] is a molecular property prediction dataset and -NO2 and -NH2 in mutagen graphs are labeled as ground-truth rationale. **AlkaneCarbonyl** [1] is a molecular property prediction dataset and alkane and carbonyl are labeled as ground-truth rationale. **FluorideCarbonyl** [1] contains 8,600 molecular graphs and benzimidazole and fluoroethyl are labeled as ground-truth rationale.

Baselines. We mainly compare with four of the latest and most competitive baselines in graph generation literature: **GDSS** [14], **DiGress** [33], **DisCo** [37] and **PARD** [42].

Setup. For filtration learning, Our backbone GNN can be instantiated with any powerful expressive self-supervised graph neural network. Since our loss relies on computational topology for graphs,

Table 1: Generation quality on rationale-dominated graphs. Top performance is in bold, runner-up is underlined.

Method	BA-2Motifs			BA-HouseAndGrid			SPMotif0.5			Benzene			Mutag		
	Rationale AUC 100.00			Rationale AUC 100.00			Rationale AUC 95.08			Rationale AUC 100			Rationale AUC 99.40		
	Deg.	Clus.	Orb.	Deg.	Clus.	Orb.	Deg.	Clus.	Orb.	Deg.	Clus.	Orb.	Deg.	Clus.	Orb.
GDSS	.046	.012	.057	.069	<u>.075</u>	.054	.253	<u>.125</u>	.072	.028	.009	<u>.004</u>	.274	<u>.0047</u>	.025
DiGress	.016	.158	.014	.023	.181	.067	.158	.101	<u>.031</u>	.884	.0007	.884	.012	.022	.002
DisCo	<u>.014</u>	.138	<u>.006</u>	<u>.015</u>	.183	.066	<u>.149</u>	.150	.025	.617	<u>.0008</u>	.018	.016	.008	<u>.0005</u>
PARD	.019	.373	.021	.027	.131	<u>.051</u>	.187	.171	.094	<u>.007</u>	.124	.009	<u>.008</u>	.0043	.0007
RARD	.005	<u>.134</u>	.002	.014	.071	.013	.125	.172	.026	.005	.154	.001	.006	.011	.0003

Table 2: Generation quality under CurFD metric. Top performance is in bold, runner-up is underlined.

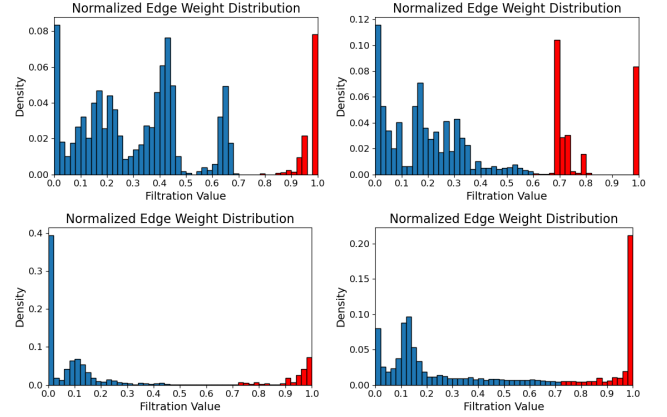
Method	BA-2Motifs	BA-HouseAndGrid	SPMotif0.5	Benzene	Mutag
GDSS	74.02	73.43	34.40	13.10	7.19
DiGress	101.50	<u>67.26</u>	<u>28.66</u>	30.70	18.56
DisCo	75.46	82.16	64.84	32.74	14.66
PARD	91.43	83.04	67.84	9.83	11.41
Ours	66.26	50.45	26.91	7.62	6.75

we employ Cellular Isomorphism Networks [10] to fully demonstrate its effectiveness. Blockwise autoregressive diffusion is also a general framework that can be combined with any equivariant network, we simply use PPGN [24] combined with transformer introduced by [42] to achieve a balance between expressivity and efficiency.

Metrics. For filtration learning, we report rationale ROC AUC [23]($\uparrow$), which measures how well the learned filtration ranks rationale edges above noise. For generic graph generation, we measure generation quality using the Maximum Mean Discrepancy (MMD) [40]($\downarrow$) as a distribution distance between the generated graphs and the test graphs, based on the distributions of **degree**, **clustering coefficient** and **occurrence count of all Orbits**. We additionally report **Curvature Filtration Distance (CurFD)** [31] ($\downarrow$), which compares graphs by measuring the distance between topological descriptors derived from their curvature filtrations. **CurFD** is provably stable, capable of counting certain substructures, and sensitive to important graph characteristics. All experiments are repeated five times, and the median result is reported.

5.2 Filtration Evaluation

We report rationale AUCs for each dataset in the headers of Tables 1 and Table 3. Across all datasets, the results exceed 95%, indicating that a topology-centered perspective effectively trains the GNN to distinguish the rationale from noise. When the rationale contains cycles (Benzene, BA-HouseAndGrid and BA-2Motifs), the rationale AUC even reaches 100%, reflecting perfect separation. We also provide the edge filtration distribution in Figure 4, filtration colored blue corresponds to G_e and the red corresponds to G_X . We can easily separate them into a wider left part and a narrower right part due to the d_{topo} .


Figure 4: Edge filtration distribution. BA-2Motifs(top left), BA-HouseAndGrid(top right), SPMotif0.5(bottom left), Benzene(bottom right)

5.3 Generic Graph Generation

Table 1 provides the generation quality of our method compared to the baselines. Our method shows outstanding performance achieving SOTA or close runner-up results, particularly in metrics of **degree** and **orbit**, while none of the baselines demonstrates consistent performance across different datasets and evaluation metrics. We additionally compare all metrics under **CurFD** (in Table 2), and our method achieves the best performance in almost all datasets.

5.4 Rationale Distribution Evaluation

To evaluate the in-distribution rationale of the generated graphs, we adopt the maximum mean discrepancy (MMD) to compare the distributions of graph statistics between the 1000 generated graphs and the ground truth rational subgraphs. Table 3 shows the MMD results on three real-world datasets. We observe that RARD outperforms the baselines in general. Especially for Benzene and FluorideCarbonyl datasets, RARD achieves the lowest MMD distances across all metrics. The MMD results verify the effectiveness of RARD in capturing the underlying distribution of rationale in the generation of rationale-dominated graphs.

Table 3: MMD distances between the generated graphs and rationales. We report MMD distances of degree distributions, cluster coefficients, spectrum distributions, and the summation. Top performance is in bold, runner-up is underlined.

Models	AlkaneCarbonyl (Rationale AUC 96.84)				Benzene (Rationale AUC 100.00)				FluorideCarbonyl (Rationale AUC 97.55)			
	Deg.	Clus.	Spec.	Sum.	Deg.	Clus.	Spec.	Sum.	Deg.	Clus.	Spec.	Sum.
GDSS	0.1292	0.0000	<u>0.0411</u>	0.1703	0.0908	0.0000	0.0394	0.1302	0.4288	<u>0.0002</u>	0.0398	0.4688
DiGress	0.1475	<u>0.0002</u>	0.0418	0.1895	0.2014	0.0018	0.0403	0.2435	0.1937	0.0000	<u>0.0396</u>	0.2333
DisCo	0.1800	<u>0.0002</u>	0.0419	0.2221	0.1916	<u>0.0003</u>	0.0403	0.2322	0.2199	0.0000	<u>0.0404</u>	0.2603
PARD	0.1122	0.0083	0.0416	<u>0.1621</u>	<u>0.0699</u>	<u>0.0026</u>	<u>0.0384</u>	<u>0.1109</u>	<u>0.1638</u>	0.0003	0.0404	<u>0.2045</u>
RARD	<u>0.1172</u>	0.0000	0.0244	0.1416	0.0530	0.0000	0.0331	0.0861	0.1006	0.0000	0.0353	0.1359

5.5 Molecular Graph Generation

We experiment with benzene in molecular graph generation task. We generate 100 molecules for stand-alone evaluation. We use the most commonly used evaluation metrics in the molecular graph literature. **Validity**($\uparrow$) is the fraction of valid molecules without valency correction. **Fr chet ChemNet Distance (FCD)**($\downarrow$) evaluates the distance between the training and generated sets using the activations of the penultimate layer of the ChemNet [19]. **Uniqueness(Uni.)**($\uparrow$) is the fraction of the valid molecules that are unique. **Neighborhood subgraph pairwise distance kernel (NSPDK)**($\downarrow$) is the MMD between the generated molecules and test molecules which takes into account both the node and edge features for evaluation. Table 4 shows that RARD surpasses the baselines on Validity and FCD, at the cost of a slight decrease in Uniqueness.

Table 4: Molecules Generation quality on Benzene

Model	Validity $\uparrow$	FCD $\downarrow$	Uni. $\uparrow$	NSPDK $\downarrow$
GDSS	95.2	14.66	99.64	0.055
DisCo	88.2	16.26	99.46	0.082
DiGress	91.0	23.06	99.25	0.064
PARD	90.6	10.99	99.73	0.082
RARD	97.0	5.98	97.99	0.019

5.6 Ablation Study

Effect of Rationale-first Ordering. We further validate the superiority of the rationale-first autoregressive ordering. Owing to the simplicity of our blockwise construction, any graph serialization order can be used as a drop-in replacement for our derived ordering. Keeping the subsequent diffusion procedure identical, we therefore introduce two alternatives—DFS ordering and random ordering—to redefine the block sequence. Additionally, we keep the rationale-first ordering unchanged while randomly permuting the block order before feeding the graph into the diffusion process. Table 5 reports the generation results on Benzene. We surprising observe that even under random permutations, the rationale-first ordering remains competitive.

Hyperparameter Study. As shown in Table 6, we conduct an ablation study on our three main hyperparameters: n (number of rationale blocks), m (number of noise blocks), and the diffusion step T . The results show that merely separating the rationale and

Table 5: Ablation on ordering and variant.

	Deg. ($\downarrow$)	Clus. ($\downarrow$)	Orb. ($\downarrow$)
Rationale-first Ordering	0.0055	0.1768	0.0012
DFS Ordering	0.0095	0.1803	0.0048
Random Ordering	0.0185	0.2332	0.0091
Rationale-first random permute	0.0057	0.1934	0.0085

noise subgraphs $(n, m) = (1, 1)$ already yields competitive performance, while introducing additional blockwise autoregressive extent $((n, m)\uparrow)$ brings further improvements. In contrast, increasing the diffusion steps per block offers only marginal gains. Consequently, we adopt $(n, m) = (2, 4)$ with 20 diffusion steps per block as the default setting for all datasets, without additional tuning.

Table 6: Ablation study on Benzene with varying (n, m) while keeping the total diffusion steps fixed (left part). The right part examines the effect of increasing steps for the fixed total number of blocks.

Setting	total steps fixed			total blocks fixed	
Total diffusion steps	120			200	400
(n,m)	1,1	2,2	2,4	2,2	2,2
diffusion steps per block	60	30	20	50	100
Deg	.003	.01	.005	.006	.006
Clus	.14	.13	.15	.22	.20
Orbit	.003	.012	.001	.011	.010
CurFD	16.57	14.76	7.62	13.40	11.55

6 Conclusion

We introduced RARD, a novel framework that exploits the rationale-noise hierarchy by maximizing the topological gap, isolating rationale edges and postponing noise. The learned filtration drives a blockwise autoregressive diffusion that samples rationale blocks first and then incrementally adds noise. RARD achieves competitive performance on seven standard datasets without using any extra features, and its rationale-first design further enhances the generation quality of the rational subgraph.

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A Variational Lower Bound Derivation

Note that while the forward process is independently applied to all elements, the backward process is coupled together with conditional independence assumption. Formally,

$$q(G_{1:T} | G_0) = \prod_{t=1}^T q(G_t | G_{t-1}, G_0),$$

$$p_\theta(G_{0:T}) = p_\theta(G_T) p_\theta(G_0 | G_1) \prod_{t=2}^T p_\theta(G_{t-1} | G_t). \quad (6)$$

$$\begin{aligned}
\log p_\theta(G_0) &= \log \int p_\theta(G_{0:T}) dG_{1:T} \\
&= \log \int q(G_{1:T} | G_0) \frac{p_\theta(G_{0:T})}{q(G_{1:T} | G_0)} dG_{1:T} \\
&\geq \mathbb{E}_{q(G_{1:T} | G_0)} \left[\log p_\theta(G_{0:T}) - \log q(G_{1:T} | G_0) \right] \\
&= \mathbb{E}_q \left[\log p_\theta(G_T) + \log p_\theta(G_0 | G_1) + \sum_{t=2}^T \log p_\theta(G_{t-1} | G_t) \right. \\
&\quad \left. - \sum_{t=1}^T \log q(G_t | G_{t-1}, G_0) \right] \\
&= -\text{KL}(q(G_T | G_0) \parallel p_\theta(G_T)) \\
&\quad + \mathbb{E}_{q(G_1 | G_0)} [\log p_\theta(G_0 | G_1)] \\
&\quad - \sum_{t=2}^T \mathbb{E}_{q(G_t | G_0)} \left[\text{KL}(q(G_{t-1} | G_t, G_0) \parallel p_\theta(G_{t-1} | G_t)) \right]. \quad (7)
\end{aligned}$$

Using (6), the first term can simplified as

$$\mathbb{E}_{q(G_1 | G_0)} \left[\sum_i \log p_\theta(v_0^i | G_1) + \sum_{i,j} \log p_\theta(e_0^{i,j} | G_1) \right], \quad (8)$$

and similarly, the t -th step loss $\mathcal{L}_t(\theta)$ is

$$\begin{aligned}
\mathbb{E}_{q(G_t | G_0)} \left[\sum_i \text{KL}(q(v_{t-1}^i | v_t^i, v_0^i) \parallel p_\theta(v_{t-1}^i | G_t)) + \right. \\
\left. \sum_{i,j} \text{KL}(q(e_{t-1}^{i,j} | e_t^{i,j}, e_0^{i,j}) \parallel p_\theta(e_{t-1}^{i,j} | G_t)) \right] \quad (9)
\end{aligned}$$

B Details of Discrete Denoising Diffusion

We closely follow DiGRESS and D3PM in setting up the forward and reverse chains for discrete diffusion, and focus on three key distributions—(a) the marginal $q(G_t | G_0)$, (b) the one-step posterior $q(G_{t-1} | G_t, G_0)$, and (c) the learnable reverse process $p_\theta(G_{t-1} | G_t)$ —that together yield an exact negative VLB loss. Because the forward process factorizes over individual variables (see (6)), it suffices to describe a single element $\mathbf{x} \in \{\mathbf{v}^i, \mathbf{e}^{i,j}\}$, each modeled as categorical, $\mathbf{x}_t \sim \text{Cat}(\mathbf{x}_t; \boldsymbol{\pi})$ with $\boldsymbol{\pi} \in [0, 1]^K$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$, so that $p(\mathbf{x}_t = \mathbf{x}_t) = \mathbf{x}_t^\top \boldsymbol{\pi}$. D3PM shows that the elementary transition $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ can be written with a matrix $Q_t \in [0, 1]^{K \times K}$ whose entries $[Q_t]_{ij} = q(\mathbf{x}_t = \mathbf{e}_j | \mathbf{x}_{t-1} = \mathbf{e}_i)$, giving

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; Q_t^\top \mathbf{x}_{t-1}).$$

Multiplying the transitions yields the t -step marginal

$$(a) \quad q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{x}_t; \bar{Q}_t^\top \mathbf{x}_0), \quad \bar{Q}_t := Q_1 \cdots Q_t,$$

while Bayes' rule gives the $(t-1)$ -step posterior

$$(b) \quad q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\mathbf{x}_{t-1}; \frac{Q_t \mathbf{x}_t \odot \bar{Q}_{t-1}^\top \mathbf{x}_0}{\mathbf{x}_t^\top \bar{Q}_t^\top \mathbf{x}_0}\right).$$

To ensure simplicity and keep $q(G_T | G_0)$ non-informative we set

$$Q_t = \alpha_t I + (1 - \alpha_t) \mathbf{1} \boldsymbol{\mu}^\top, \quad \alpha_t \in [0, 1],$$

where $\boldsymbol{\mu}$ is uniform ($1/K_v$ for nodes, $1/K_e$ for edges; DiGRESS instead uses empirical marginals). Since the true reverse process

satisfies $p(\mathbf{x}_{t-1} | \mathbf{x}_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t)$, we parameterize

$$(c) \quad p_\theta(\mathbf{x}_{t-1} | G_t) = \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) p_\theta(\mathbf{x}_0 | G_t),$$

letting a neural network output $p_\theta(\mathbf{x}_0 | G_t)$ directly. Combining (a), (b) and (c) yields an exact, element-wise tractable negative VLB objective for training the discrete diffusion model.

C Bottleneck Distance

Definition 1 (Bottleneck Distance). *Let P_1, P_2 be two persistent barcodes. A partial matching $\pi \subseteq P_1 \times P_2$ pairs each point in P_1 and P_2 with at most one partner. For any $p = (b, d) \in \mathbb{R}^2$ set $\bar{p} = (d - b, d - b)$, the closest point of p to the diagonal $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$. The bottleneck distance is then*

$$\begin{aligned}
d_B(P_1, P_2) &= \inf_{\pi} \max \left\{ \max_{(p,q) \in \pi} \|p - q\|_\infty, \right. \\
&\quad \max_{p \in P_1 \setminus \pi_1} \|p - \bar{p}\|_\infty, \\
&\quad \left. \max_{q \in P_2 \setminus \pi_2} \|q - \bar{q}\|_\infty \right\}. \quad (10)
\end{aligned}$$

where $\|\cdot\|_\infty$ is the ∞ -norm and π_1, π_2 are the projections of π onto P_1, P_2 .

Intuitively, d_B is the minimal cost of morphing one barcode into the other: matched points may move toward each other, while unmatched points may slide onto the diagonal, and the cost of a plan is the largest ∞ -norm displacement required. It is shown that the bottleneck distance is a stable metric in persistent barcodes space, which means that small perturbations in the input data will not change the bottleneck distance too much.

D Missing Proof for Theorem 1

PROOF. Step 1: A computable lower bound to d_{topo} . By Kantorovich–Rubinstein duality and the L_ψ -Lipschitz property of ψ ,

$$d_{\text{topo}}(\mathcal{D}(G^{\leq \lambda^*}), \mathcal{D}(G^{\geq \lambda^*})) \geq \frac{1}{L_\psi} \left\| \mathbb{E}[T_X] - \mathbb{E}[T_\varepsilon] \right\|. \quad (11)$$

Hence maximizing the empirical feature separation $\|\widehat{\mathbb{E}}[T_X] - \widehat{\mathbb{E}}[T_\varepsilon]\|$ maximizes a computable lower bound to the true topological gap.

Step 2: From diagram margin to edge-score separation. Let f° be a filtration in the hypothesis class that achieves the population supremum of the RHS in (11). By (A2),

$$\left\| \mathbb{E}[T_X(f^\circ)] - \mathbb{E}[T_\varepsilon(f^\circ)] \right\| \geq L_\psi \Delta.$$

Uniform convergence of the Lipschitz CE estimators implies, with probability at least $1 - \eta$,

$$\left\| \mathbb{E}[T_X(\hat{f}_\alpha)] - \mathbb{E}[T_\varepsilon(\hat{f}_\alpha)] \right\| \geq L_\psi \Delta - \epsilon_{\text{opt}} - \epsilon_{\text{gen}}(N). \quad (*)$$

Next, relate the feature gap to *edge-score separation*. Persistence stability for sublevel filtrations gives, for any two edge-weight functions w, w' on E ,

$$d_B(\mathcal{D}(w), \mathcal{D}(w')) \leq \|w - w'\|_\infty.$$

Consider perturbing the learned filtration $\hat{f}_\alpha$ so that rationale and noise edges are *not* separated (i.e., their score distributions overlap substantially); then $G^{\leq \lambda}$ and $G^{\geq \lambda}$ interleave many of the same

edges, forcing their diagrams to be close in d_B and, via (11), forcing the feature gap to shrink. Conversely, a positive diagram margin implies that, on average, rationale edges must receive larger filtration values than noise edges by at least $\kappa \Delta$ for some $\kappa > 0$ depending only on the diagram family and ψ (this follows from a coupling argument that counts the number of critical events—component merges and cycle closures—whose birth/death times move by at least the edge-score margin). Combining this with $(\star)$ establishes (C1), with the additive slack absorbed into $\epsilon_{\text{opt}} + \epsilon_{\text{gen}}(N) + \epsilon_{\text{suf}}$.

Step 3: Binary convergence in probability. Let $\bar{c}_X = \mathbb{E}[f_{\hat{\alpha}}(e_X)]$ and $\bar{c}_\epsilon = \mathbb{E}[f_{\hat{\alpha}}(e_\epsilon)]$. The objective in $(\star)$ encourages the means to separate while the Lipschitz constraint discourages oscillatory solutions. By standard uniform convergence and compactness, any sequence of near-optimizers admits a subsequence for which the intra-class variances $\text{Var}(f_{\hat{\alpha}}(e_X))$ and $\text{Var}(f_{\hat{\alpha}}(e_\epsilon))$ go to zero (otherwise the lower bound cannot approach its supremum under a fixed Lipschitz constant). Therefore, $f_{\hat{\alpha}}(e_X) \xrightarrow{P} \bar{c}_X$ and $f_{\hat{\alpha}}(e_\epsilon) \xrightarrow{P} \bar{c}_\epsilon$, and (C1) implies $\bar{c}_X > \bar{c}_\epsilon$. Any strictly increasing reparameterization of $[0, 1]$ yields an equivalent filtration, proving (C2).

Finally, assumption (A1) guarantees that small residual dependence on E_ϵ (bounded by ϵ_{suf}) does not alter the ordering conclusion, only the constants. Collecting the steps completes the proof. $\square$

E Data Detailed

The statistics and properties of the dataset are as follows:

Table 7: Statistics and properties of datasets. The datasets above the dashed line are synthetic, and those below are real-world.

Dataset	# Graphs	# CIs	Avg. $ V $	Avg. $ E $
BA-2Motifs	1 000	2	25.00	25.48
SPMotif	18 000	3	45.98	66.72
BA-HouseAndGrid	10 000	2	29.42	41.01
Mutag	2 951	2	30.13	30.45
Benzene	12 000	2	20.58	21.83
AlkaneCarbonyl	1 125	2	21.40	21.59
FluorideCarbonyl	8 671	2	21.42	21.62

F Complexity Analysis

F.1 PH Complexity

For H0/H1 persistent homology on a graph with m edges and n vertices, we practically follow the link-cut-tree based algorithm, resulting in $O(m \log n)$ depth and $O(mn)$ work parallel algorithm. On our datasets $m \leq 70$, so PH is small compared with the PPGN-Transformer.

F.2 Generation Complexity

The asymptotic runtime complexity for sampling a graph with N nodes from an RARD model with T total steps is :

$$O(T^2 N + T N^3)$$

For sampling with RARD on a graph of N nodes over T denoising steps, step t operates only on the generated prefix of size $n_t \leq N$.

Algorithm 1 Joint Training of Block-Size Distribution and Block-wise Denoising Diffusion

Require: Graph $G = (V, E)$, filtration scores $f_{\hat{\alpha}} : E \rightarrow [0, 1]$, diffusion steps T , predictors g_θ (block size) and f_θ (block denoiser)

- 1: // **Block construction**
- 2: Threshold edges at τ to get E_X, E_ϵ ; build V_X, V_ϵ and coarse blocks $B^{(0)} = \{V_X, V_\epsilon\}$
- 3: Compute node scores $s(v)$, sort within each coarse block, and split into $\{B_1^X, \dots, B_n^X\}$ and $\{B_1^\epsilon, \dots, B_m^\epsilon\}$
- 4: Concatenate to obtain $B = \{B_1, \dots, B_L\}$
- 5: // **Blocksize prediction loss (for $p_\theta(|B_i| | G[B_{<i}])$)**
- 6: Set $|B_{L+1}| = 0$
- 7: **for** $i = 1$ **to** L **do**
- 8: $\ell_i^{\text{size}} \leftarrow \text{CrossEntropy}(g_\theta(G[B_{<i}]), |B_i|)$
- 9: **end for**
- 10: // **Blockwise diffusion (for $p_\theta(x | G[B_{<i}] \cup \emptyset[B_i])$)**
- 11: Sample a diffusion step $t \sim \mathcal{U}\{1, \dots, T\}$
- 12: **for** $t = 1$ **to** T **do**
- 13: $M \leftarrow \text{index mask of } G[B_{\leq t}] \setminus G[B_{<t}]$
- 14: Sample a noise graph $G[\tilde{B}_{\leq t}] \sim q_{t|0}(G[B_{\leq t}])$
- 15: $G[\tilde{B}_{\leq t}] \leftarrow M \odot G[\tilde{B}_{\leq t}] + (1 - M) \odot G[B_{\leq t}]$
- 16: $X \leftarrow f_\theta(G[\tilde{B}_{\leq t}]) \odot M$
- 17: $Y \leftarrow G[B_{\leq t}] \odot M$
- 18: $\ell_t^{\text{diff}} \leftarrow \mathcal{L}_t^{\text{CE}}(X, Y)$ (Eq.1)
- 19: **end for**
- 20: **Minimize** $\sum_{i=1}^L \ell_i^{\text{size}} + \sum_{i=1}^L \ell_i^{\text{diff}}$

The per-step *spatial* cost is $O(n_t^p)$ (with $p = 2$ for a Transformer-only backbone and $p = 3$ for a PPGN-Transformer backbone). While the *temporal* term reflects conditioning on past states within a block, at denoising step t each of the n_t currently materialized nodes may attend to (or retrieve from) up to the t previous step-wise representations of that same prefix (e.g., causal temporal attention, cached message states, or per-step residual memories). Hence a conservative per-step bound is $O(n_t t)$. Aggregating across steps gives

$$\sum_{t=1}^T n_t t \leq \sum_{t=1}^T N t = N \cdot \frac{T(T+1)}{2} = O(T^2 N),$$

where we use $n_t \leq N$ (prefix growth) and the closed form for $\sum_{t=1}^T t$. For the spatial term, $\sum_{t=1}^T n_t^p \leq \sum_{t=1}^T N^p = T N^p$ since $n_t \leq N$ for all t . Combining the two pieces yields

$$\sum_{t=1}^T (n_t t + n_t^p) \leq \sum_{t=1}^T (N t + N^p) = O(T^2 N + T N^p),$$

a worst-case upper bound; in practice, block resets cap the effective history and the monotone growth $n_t \uparrow$ further reduces constants.

Conclusion: the overall runtime is $O(T^2 N + T N^2)$ with a Transformer-only backbone and $O(T^2 N + T N^3)$ with a PPGN-Transformer backbone; the bound is conservative since $n_t \leq N$.

G Visualization

As shown in Figure 5, our rationale-first filtration can effectively distinguish between rationale subgraphs and noise subgraphs. When rationale is a simple case, RARD remains effective. Our experimental dataset contains such situations (Mutag, AlkaneCarbonyl), and even if rationale has only two edges or a simple molecular chain, RARD can still stably identify it.

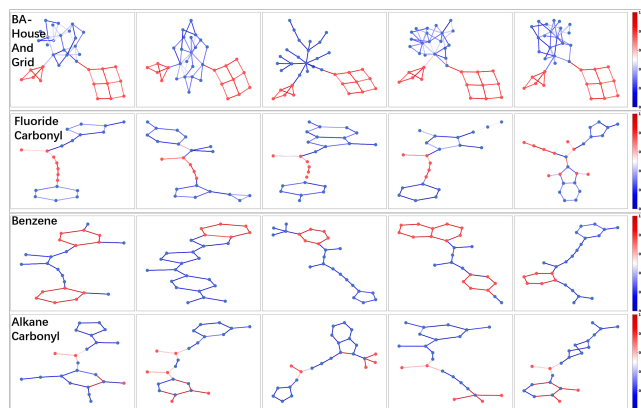


Figure 5: Visualization of the learned graph filtrations. Nodes colored red correspond to ground-truth rational nodes.